STATISTICS FOR ECONOMISTS · LESE152

Practice Set 3

CHALLENGE / 심화 | 여러 조건과 공식을 연결하여 응용하는 단계

Name: _____ **Student ID:** _____

Exam structure: 10 MCQ + 10 T/F + 3 short-answer questions Total: 100 points | Suggested time: 80-95 min

Instructions: Choose one answer per MCQ. Mark True/False and provide a brief explanation. Show all major calculation steps in Part C. Check units, conditioning events, and independence assumptions.

Standard normal: $\Phi(0.5)=0.6915$, $\Phi(1)=0.8413$, $\Phi(1.2816)=0.9000$, $\Phi(1.5)=0.9332$, $\Phi(1.645)=0.9500$, $\Phi(1.96)=0.9750$, $\Phi(2)=0.9772$.

For the standard normal, $\Phi(-z)=1-\Phi(z)$. Useful exponentials: $e^{-1}=0.3679$, $e^{-2}=0.1353$, $e^{-3}=0.0498$, $e^{-4}=0.0183$.

Percentile position convention: $Lp=(p/100)(n+1)$, with linear interpolation between neighboring sorted observations.

PART A · MULTIPLE CHOICE (10 × 2 = 20 points)

MC1. Waiting time T is exponential with rate 0.25 per minute. Find $P(T > 8)$.

(a) 0.0183 (b) 0.1353
(c) 0.3679 (d) 0.8647

MC2. $P(A)=0.3$, $P(B)=0.4$, $P(A \cap B)=0.12$. Find $P(B \mid A)$.

(a) 0.12 (b) 0.30
(c) 0.40 (d) 0.70

MC3. X and Y are independent; sd(X)=3 and sd(Y)=4. Find Var(2X-Y).

(a) 14 (b) 25
(c) 52 (d) 100

MC4. X ~ Binomial(6, 0.5). What is $P(X \geq 5)$?

(a) 1/64 (b) 6/64
(c) 7/64 (d) 15/64

MC5. Arrivals are Poisson with rate 2 per 30 minutes. Find P(no arrivals in 1 hour).

(a) e^{-1} (b) e^{-2}
(c) e^{-3} (d) e^{-4}

MC6. X ~ Uniform(2,6). Find $P(3 < X < 5)$.

(a) 1/4 (b) 1/2
(c) 2/3 (d) 3/4

MC7. P(Good weather)=0.2, E(Y|Good)=30, E(Y|Bad)=10. Find E(Y).

(a) 12 (b) 14

(c) 18 (d) 20

MC8. A long right tail in the distribution of income most commonly suggests:

(a) Mean < median (b) Mean > median

(c) Mean = 0 (d) Variance = 0

MC9. Disease prevalence is 0.05, sensitivity 0.90, and false-positive rate 0.05. Find P(disease | +).

(a) 0.0500 (b) 0.4500

(c) 0.4865 (d) 0.9000

MC10. $Z=(X-\mu)/\sigma$, where $0<\sigma<\infty$. What is sd(Z)?

(a) 0 (b) 1

(c) σ (d) σ^2

PART B · TRUE / FALSE (10 × 3 = 30 points)

TF1. Zero covariance between X and Y always proves that they are independent.

Mark: True [ ] False [ ] Explanation: _____

TF2. The identity E(X+Y)=E(X)+E(Y) requires X and Y to be independent.

Mark: True [ ] False [ ] Explanation: _____

TF3. For exponential T, P(T>30 | T>10) = P(T>20).

Mark: True [ ] False [ ] Explanation: _____

TF4. For a continuous random variable, P(X=3) may be positive.

Mark: True [ ] False [ ] Explanation: _____

TF5. For any random variable X with finite variance, Var(2X-7)=4Var(X).

Mark: True [ ] False [ ] Explanation: _____

TF6. A positive sample correlation is sufficient evidence that changing X will causally increase Y.

Mark: True [ ] False [ ] Explanation: _____

TF7. If A is a subset of B and P(B)>0, then P(A|B)=P(A)/P(B).

Mark: True [ ] False [ ] Explanation: _____

TF8. If X ~ Poisson(4), P(X=0)=e^(-4).

Mark: True [ ] False [ ] Explanation: _____

TF9. If the CDF F is continuous at x=3, then P(X=3)=0.

Mark: True [ ] False [ ] Explanation: _____

TF10. If X and Y are independent with finite variances, Var(X-Y)=Var(X)+Var(Y).

Mark: True [ ] False [ ] Explanation: _____

PART C · SHORT ANSWERS (50 points)

C1. Rare high-sales day

Daily farm-store sales (in tens of thousand won), sorted: 11, 12, 13, 13, 15, 16, 18, 19, 21, 48.

(a) [5 pt] Calculate the mean and median and explain the effect of the large observation.

(b) [5 pt] Use Lp=(p/100)(n+1) to find Q1, Q3, and the upper outlier fence.

(c) [5 pt] Calculate the sample standard deviation and CV. Explain why CV can be high here.

C2. A conflicting second diagnostic test

Only 5% of farms carry a crop disease D. Each test has sensitivity P(+|D)=0.95 and false-positive rate P(+|D^c)=0.08. The two test results are conditionally independent given D or D^c.

(a) [5 pt] Find the probability that the first test is positive.

(b) [5 pt] Given one positive test, find P(D|+).

(c) [5 pt] The first test is positive but the second test is negative. Find P(D|+,-).

C3. Risky farm returns and independent cost

Per-acre random return X (thousand won) has PMF: x = -20, 0, 40, 100 with probabilities 0.2, 0.3, 0.4, 0.1.

(a) [5 pt] Calculate E(X), E(X^2), Var(X), and sd(X).

(b) [5 pt] Let Y=2X-15. Calculate E(Y) and sd(Y).

(c) [5 pt] An independent variable cost C has E(C)=10 and sd(C)=12. For Z=2X-C-15, find E(Z) and sd(Z).

(d) [5 pt] Delivery time T ~ Normal(40, 6^2) minutes. Find P(T>49) and the 95th percentile.

Answer Key & Detailed Solutions

정답 확인은 문제를 먼저 끝까지 풀고 진행하세요. 각 해설에는 공식과 계산의 근거를 포함했습니다.

PART A · ANSWERS & EXPLANATIONS

MC1 B MC2 C MC3 C MC4 C MC5 D MC6 B MC7 B MC8 B MC9 C MC10 B

MC1 · B 지수분포 생존함수 $P(T>t)=e^{(-\lambda t)}$; $\lambda t=0.25\times 8=2$이므로 $e^{(-2)}\approx 0.1353$.

MC2 · C $P(B|A)=P(A\cap B)/P(A)=0.12/0.3=0.4$. 이는 $P(B)$와 같으므로 독립이다.

MC3 · C $\mathrm{Var}(2X-Y)=2^2\mathrm{Var}(X)+(-1)^2\mathrm{Var}(Y)=4\times 9+16=52$. 공분산은 독립이므로 0이다.

MC4 · C $P(X=5)+P(X=6)=[C(6,5)+C(6,6)](0.5)^6=(6+1)/64=7/64$.

MC5 · D 1시간은 30분의 2배이므로 $\lambda=4$. $P(X=0)=e^{(-4)}\approx 0.0183$.

MC6 · B 연속균등분포에서 $(5-3)/(6-2)=2/4=0.5$. 경계점의 포함 여부는 확률에 영향을 주지 않는다.

MC7 · B 전기댓값법칙: $E(Y)=0.2\times 30+0.8\times 10=6+8=14$.

MC8 · B 일반적으로 큰 고소득 값이 평균을 오른쪽으로 끌어올리므로 평균이 중앙값보다 크다.

MC9 · C 베이즈: $0.05\times 0.90/[0.05\times 0.90+0.95\times 0.05]=0.045/0.0925\approx 0.4865$.

MC10 · B $\mathrm{Var}(Z)=\mathrm{Var}((1/\sigma)X-\mu/\sigma)=\mathrm{Var}(X)/\sigma^2=1$. 따라서 표준편차도 1이다.

PART B · ANSWERS & EXPLANATIONS

TF1 · FALSE 공분산 0은 선형 연관이 없음을 뜻할 수 있으나 비선형 의존성이 남을 수 있다. 독립이면 공분산 0이지만 역은 일반적으로 거짓이다.

TF2 · FALSE 기댓값의 선형성은 독립 여부에 상관없이 성립한다(기댓값 존재 가정).

TF3 · TRUE 무기억성: $P(T>s+t|T>s)=P(T>t)$이며 여기서는 $s=10$, $t=20$이다.

TF4 · FALSE 연속형 확률변수에서 정확히 한 점에 있을 확률은 0이다.

TF5 · TRUE 상수 이동 -7은 분산에 영향이 없고 계수 2는 제곱되어 4배가 된다.

TF6 · FALSE 상관관계는 제3의 요인이나 역인과 가능성을 배제하지 않는다.

TF7 · TRUE $A\subseteq B$이면 $A\cap B=A$이므로 조건부확률 정의에 의해 바로 성립한다.

TF8 · TRUE 포아송 PMF에서 $x=0$을 대입하면 $e^{(-4)}\times 4^0/0!=e^{(-4)}$이다.

TF9 · TRUE $P(X=3)=F(3)-F(3-)$, 즉 CDF의 해당 점 점프 크기다. 연속이면 점프가 없다.

TF10 · TRUE $\mathrm{Var}(X-Y)=\mathrm{Var}(X)+\mathrm{Var}(Y)-2\mathrm{Cov}(X,Y)$. 독립이면 공분산 0이다.

PART C · STEP-BY-STEP SOLUTIONS

C1. Rare high-sales day

(a) 합계 186, n=10이므로 평균=18.6. 중앙값=(15+16)/2=15.5. 큰 값 48이 평균을 끌어올리므로 평균이 중앙값보다 크다.

(b) L25=0.25×11=2.75→Q1=12+0.75(13-12)=12.75. L75=0.75×11=8.25→Q3=19+0.25(21-19)=19.5. IQR=6.75; 상한=19.5+1.5×6.75=29.625. 48은 명백한 잠재적 이상치다.

(c) $\Sigma(x_i-18.6)^2=1054.4$. $s^2=1054.4/9\approx117.1556$, $s\approx10.8238$, CV=(10.8238/18.6)×100≈58.19%. 48이라는 큰 관측값이 제곱편차를 크게 만들어 산포가 커졌다.

C2. A conflicting second diagnostic test

(a) P(+)=0.05×0.95+0.95×0.08=0.0475+0.076=0.1235.

(b) P(D|+)=0.0475/0.1235≈0.3846. 양성이라도 기저유병률이 낮아서 사후확률이 약 38.5%다.

(c) 조건부 독립성으로 P(+ ,-|D)=0.95×0.05, P(+ ,-|D^c)=0.08×0.92. 따라서 P(D|+,-)=0.05×0.95×0.05/[0.05×0.95×0.05+0.95×0.08×0.92]=0.002375/(0.002375+0.06992)≈0.0329. 두 번째 음성 결과가 사후확률을 크게 낮춘다.

C3. Risky farm returns and independent cost

(a) E(X)=-20×0.2+0×0.3+40×0.4+100×0.1=-4+16+10=22. $E(X^2)=400\times.2+1600\times.4+10000\times.1=80+640+1000=1720$. Var(X)=1720-22²=1236; sd(X)=√1236≈35.1568.

(b) E(Y)=2×22-15=29. sd(Y)=2√1236≈70.3136(천 원). 상수 -15는 표준편차에 영향을 주지 않는다.

(c) E(Z)=2×22-10-15=19. X와 C가 독립이므로 Var(Z)=2²×1236+(-1)²×12²=4944+144=5088. sd(Z)=√5088≈71.3302.

(d) Z표준화=(49-40)/6=1.5 → P(T>49)=1-Φ(1.5)=0.0668. 95백분위수=40+1.645×6=49.87분.

FINAL FORMULA & ERROR CHECK

Descriptive	Sample variance $s^2=\Sigma(x_i-\bar{x})^2/(n-1)$; $sd=\sqrt{s^2}$; CV=sd/x-bar ×100%; IQR=Q3-Q1.
Probability	$P(A\cup B)=P(A)+P(B)-P(A\cap B)$; $P(A\|B)=P(A\cap B)/P(B)$.
Bayes	P(D|+)=P(+|D)P(D)/[P(+|D)P(D)+P(+|D^c)P(D^c)].
Moments	$E(X)=\Sigma x\,p(x)$; $Var(X)=E(X^2)-[E(X)]^2$; E(a+bX)=a+bE(X); $Var(a+bX)=b^2Var(X)$.
Distributions	Binomial: E=np, Var=np(1-p). Poisson: $P(X=k)=\exp(-\lambda)\times\lambda^k/k!$, E=Var=λ.
Continuous	Uniform(a,b): P(c<X<d)=(d-c)/(b-a). Exponential(rate λ): P(T>t)=exp(-λt).

마지막 점검: ① 확률의 합 1 확인 ② 조건부확률의 분모 재확인 ③ n / n-1 구별 ④ $E(X^2)$와 $[E(X)]^2$ 구별 ⑤ 분산과 표준편차의 단위 구별 ⑥ 독립성 가정 확인